

Hooman Ramezani

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EDUCATION

University of Toronto <i>MASc in Machine Learning & Operations Research</i>	Toronto, ON 2023 – 2025
- Thesis: Building healthcare LLM and ViT models with multimodal clinical data for lung cancer treatment - GPA: 4.0 / 4.0 — Coursework: Large Language Models, Cloud Data (Spark, AWS), Deep Learning Theory	
University of Waterloo <i>BASc in Systems Design Engineering</i>	Waterloo, ON 2018 – 2023
GPA: 3.7 / 4.0 — Coursework: Intro. Deep Learning, Intro. Machine Learning, Pattern Recognition, Neuroscience	

EXPERIENCE

Machine Learning Researcher <i>University Health Network</i>	May 2023 – Present FT – Toronto, ON
- First-author ‘LN-Transformer’ paper for lung nodule segmentation, achieving DiceC of 94.2% (SOTA) - Built multi-modal LLM with unstructured text, CT slices for treatment recommendations with 84% concordance	
Machine Learning Engineer <i>Advanced Micro Devices (AMD)</i>	Sept. 2022 – April 2023 Intern – Toronto, ON
- Optimized LLM inference on AMD CPUs via QAT, post-quantization, pruning, and knowledge distillation - Deployed on Triton Inference Server with TensorRT backend, leveraging dynamic batching and optimized GPU kernel execution to reduce inference latency by 80% and maximize throughput - Scaled multi-node and GPU training via PyTorch DDP and model/data parallelism (FSDP2) to large clusters	
Machine Learning Engineer <i>DarwinAI (Acquired by Apple)</i>	Jan. 2022 – April 2022 Intern – Remote
- Delivered medical vision model to Pfizer for advanced liver fibrosis diagnosis, achieving 91% sensitivity - Built end-to-end ML solutions to clients with MLOps CI/CD pipelines, enabling continuous retraining, monitoring	
Computer Vision Engineer <i>Applied Brain Research</i>	May 2021 – Aug. 2021 Intern – Waterloo, ON
- Delivered 94% accurate drone object detection edge-deployed model to BP for remote rust detection - Designed Unreal Engine 4 simulation to synthetically generate 50,000 annotated images to overcome lack of data	

RESEARCH // PROJECTS

Livy Education Chatbot Agent <i>GPT-4, RAG, LangChain, Chroma</i>	2024
- Developed a RAG-powered AI chatbot for Canvas students, integrating GPT-4 for assignment research assistance - Utilized Chroma vector database and fine-tuned embeddings for course material reranking, improving relevancy - Built context-aware AI agent to adaptively monitor student engagement and provide real-time support	
Advanced Detection of Parkinson’s Freezing of Gait <i>CNN, Time-Series Data</i>	2023
- Achieved 94% accuracy for fall forecasting for early detection of Freezing of Gait (FOG) in Parkinson’s patients - Built novel preprocessing techniques for EMG, ECG sensor data, improving time-series feature extraction	
Enhanced Robotic Grasping with PointNet <i>LiDAR, Robotics, 3D Computer Vision</i>	2022
Trained grasp proposition neural-net achieving 87% grasp success rate, adapting custom PointNet deep learning models to identify optimal grasp points from LiDAR camera, on dataset with 50k+ annotated grasp scenarios	

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL, JavaScript
ML & DL: PyTorch, TensorFlow, Hugging Face, LoRA / PEFT, RAG, AI Agents, TensorRT, ONNX, Triton, CUDA
Distributed Systems: PyTorch DDP, FSDP, Spark, Kafka, Airflow, Docker, Kubernetes, Slurm
Cloud & MLOps: AWS (S3, EC2, Lambda, SageMaker), GCP, MLflow, Prometheus, Redis, Flask, Git